Clinical Decision Support System for Disease Risk Assessment

Progress Report - 1

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Course: **CSIS 4495 – Applied Research Project**

Section: **03**

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1. Work Date / Hours Log

Date	Hours	Description of Work Done
Jan 27, 2026	2.5	Considered PDF extraction, Tesseract OCR and Streamlining documentation before implementation.
Jan 29, 2026	3	Created a Virtual Environment for the implementation in VS Code, Installed necessary Libraries for Tesseract OCR , Implemented PDF text extraction and OCR pipeline using a standard CVC report;
Jan 31, 2026	2	Coordinated with Uthara to integrate preprocessing logic into extraction outputs from the PDF and edited the code to remove irrelevant characters
Feb 2, 2026	2.5	Resolved path and import configuration related issues, Coordinated outputs for preprocessing the extracted data;
Feb 4, 2026	1.5	Research (Disease Indicators): 'Stroke' and 'Diabetes' indicators from patient-accessible reports
Feb 8, 2026	2.5	Tried to learn Streamlit UI structure, Reviewed the prototype made by Uthara, Discussed & Provided some feedback to her to make the interface better & user friendly;

Summary Description of work done during reporting period:

During Weeks 4–5, I focused primarily on implementing the core medical report in PDF format and OCR extraction pipeline, which is the foundation of the Clinical Decision Support System. Before starting the implementation in VS Code software, I went through documentation and samples of PDF extraction libraries in Python, Tesseract OCR, and Streamlit to verify their viability and to make sure that I used the right tools and libraries. The input was done by processing text-based PDFs and scanned medical reports. Path and import configuration was one of the initial problems and is especially encountered when running Python scripts over environments. This was solved through the standardization of file paths and import references. The other problem that was found during testing was that the initial extraction results included the entire text of the report, complete with multiple headers, footers, notes, and irrelevant characters. Besides this, I tried to ensure that I provided some assistance to Uthara on the Streamlit interface, such as inspecting the structure of the UI, testing the upload flow, and ensuring that the interface was reflective of the post-extraction pipeline. At this point, the legitimacy of extraction was confirmed to be through a single medical report sample, and refined before scaling. Simultaneously, I also carried out a study on stroke and diabetes and listed the indicators they reported commonly to aid in future feature selection for feeding into the Machine Learning models.

Repo Check-In :

I tried to release the entire PDF and OCR extraction implementation to the GitHub repository. It contains scripts and a configuration to process text-based PDFs and scanned medical/CVC report by using Tesseract OCR and additional refinements to remove irrelevant text of the report. I incorporated the logic of preprocessing that Uthara offers in the extraction workflow and made sure to match the extraction outputs and downstream processing. The code on all extraction-related issues was posted by me, and the UI based on Streamlit was posted by Uthara.

The files/folders I have checked in the repo as follows:

1. Implementation/app.py

2. AI Prompts

1. PDF & OCR Extraction Workflow: "*How to combine PDF text extraction and Tesseract OCR to retrieve the data in text-based and scanned medical reports?"*

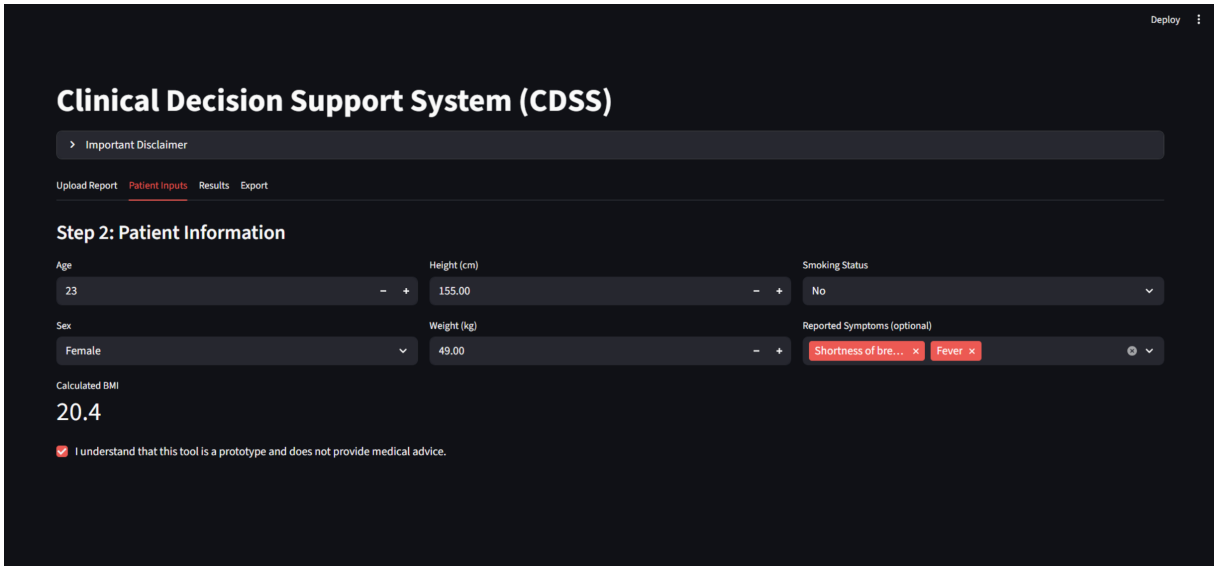
2. Handling Noisy Medical PDF Output: "*What are the typical difficulties associated with this process of extracting data out of medical PDF reports, and how can I filter out irrelevant characters?"*

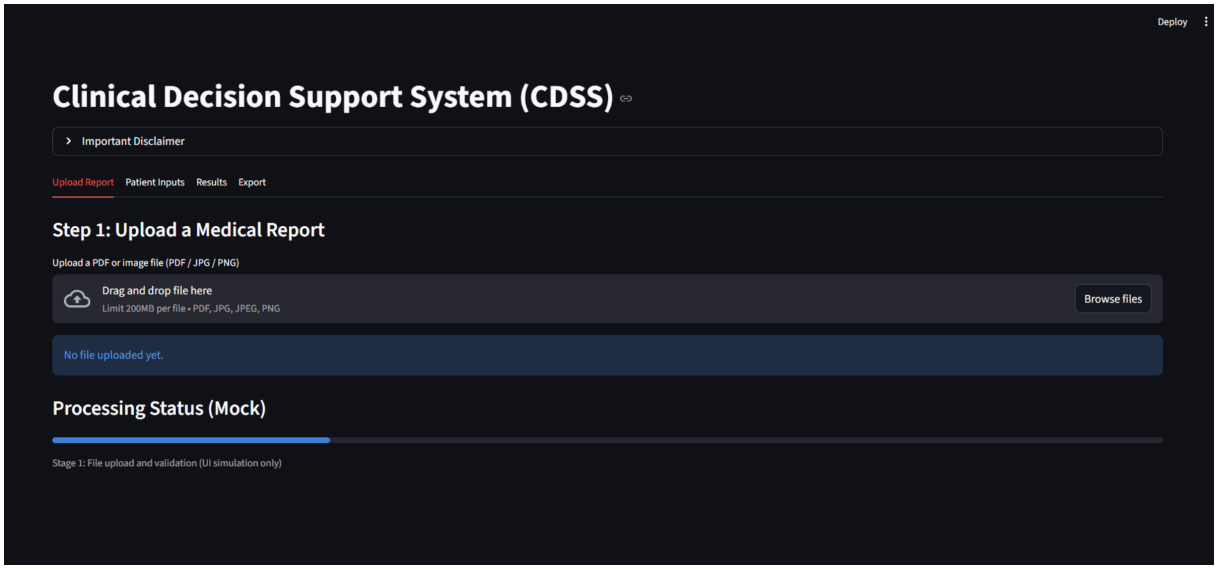
3. Disease Indicator Research (Stroke & Diabetes): *"What clinical factors/features are commonly present in laboratory reports and summaries for assessing stroke and diabetes risk? Suggest to me the existing research papers on this"*

AI Tool	**Version / Account**	**Purpose of Use**	**Value Added**
ChatGPT	GPT-5.2 (Premium)	Conceptual clarification of PDF extraction and OCR workflows	Implementation of the PDF/OCR extraction pipeline
ChatGPT	GPT-5.2 (Premium)	Understanding challenges in extracting relevant data from medical PDFs	Modification of the extraction logic to remove irrelevant report content and mark down useful indicators

ChatGPT	GPT-5.2 (Premium)	Support for disease-indicator research (Stroke and Diabetes)	Identification of relevant indicators to support future feature selection and support Machine Learning models

3. User Interface Screenshots:





File change. Rerun Always rerun

Upload Report Patient Inputs Results Export

Step 3: Results Overview (Mock Data)

Extracted Laboratory Indicators

	Indicator	Value	Unit	Reference Range	Flag
0	Hemoglobin	15	g/dL	13–17	Normal
1	Total Leukocyte Count	5100	/cumm	4800–10800	Normal
2	Lymphocyte Percentage	18	%	20–40	Low
3	MCHC	35.7	%	31.5–34.5	High

Overall Risk Indicator (Mock)

Risk Score

0.32

Model not implemented yet (planned future stage)

Explanation Summary

- Lymphocyte percentage flagged as low
- MCHC flagged as high
- Risk score shown here is illustrative only

Processing Progress (Mock)

Stage 2: Structured data extraction (planned) • Stage 3: Model-based risk scoring (planned)